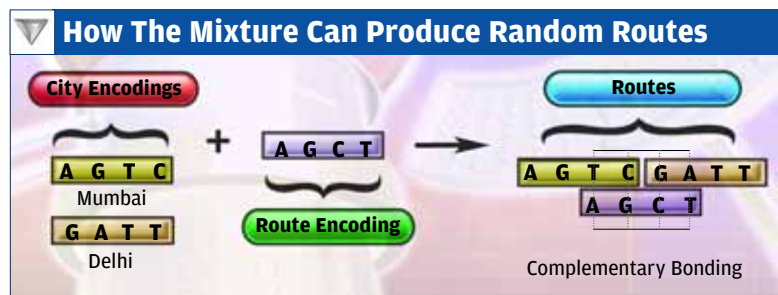


finally filter out the molecules in the test tube. Those DNA molecules that emerged gave him the answer to the seven-city version of the TSP.

If that sounds way out, it's only because we don't understand how advanced biotechnology is today. It's relatively easy to understand the 'hardware' part of it—that DNA molecules, with A, C, G and T, play around and recombine and split and so on. What seems difficult to understand is how they play around the way we want them to. It turns out that using the appropriate biochemical techniques, one can 'tell' the DNA molecules what to do, and one can also 'read out' the results of what went on in the test tube.

It is beyond our scope here to explain *how* exactly these biochemical agents work; what we can do is explain *what* manipulations are done. And what better example than Adleman's experiment?

To understand how a mathematical problem



We can obtain the route encodings from the city encodings. These two mixed together result in complementary bonding—producing molecules that represent routes

can be solved by a bubbling solution in a scientist's test tube requires us to understand a few things. First, the structure of the molecules in the solution. Second, what goes on inside the test tube and how. And third, how the scientist gets his answers from the test tube.

Now let's get our hands dirty and see how Adleman's experiment worked.

Adleman's Experiment (Or Something Like It)

DNA molecules consist, like we said, of the four nucleotides A, T, G and C. These can be considered its building blocks. The familiar double-helix you see in all the pictures comes about this way: A (Adenine) bonds with T (Thiamine), and C (Cytosine) bonds with G (Guanine), to produce what is called a complementary strand. So if you have a strand like ACCTCA, the complement would be TGGAGT. Those two strands entwine in double-helix fashion.

We'll restrict the problem a little so there's only one correct answer. (Let's take it easy on Adleman, shall we?) So let's say there are no flights between some cities, such that the *only* answer is M-D-B-C-K. So what we hope to achieve with this DNA computation is DNA strands that encode this particular route.

Our first step would, naturally, be to encode the cities and the routes. Let's use four bases to encode a city, so Mumbai could be, say, AGTC, Delhi could be GATT, and so on. Our first task is to produce a test tube full of these molecules. That's easy to do with today's techniques—all that's needed is a DNA synthesiser. (OK, we're skipping a lot of details, but those aren't important.)

Now how would we produce molecules that encode the routes between cities? We use a property of DNA for this—the fact that a strand of DNA entwines ('hybridises') with its complementary sequence. So, suppose we took the last two bases of Mumbai (TC), and the first two bases of Delhi (GA). We get TCGA. Now we take the complement of this, which is AGCT. This can represent the path between Mumbai and Delhi. How? Because AGCT would hybridise with TCGA, linking the two strands together.

So if we had lots of copies of Mumbai (AGTC) and Delhi (GATT), and lots of copies of the route between them (AGCT), we'd get several molecules that strung together Mumbai and Delhi. The molecules that string together Mumbai and Delhi would represent the route between them.

The same holds for more than two cities—you could have routes consisting of any number of cities. After a sufficient period of time, if you'd taken enough DNA in the test tube, you'd have all the paths possible—including paths that make no sense for our purposes, such as Mumbai-Bangalore-Delhi. These answers are nonsensical because we're assuming there's only one answer to the problem.

We need to weed out the paths that make no sense, so our next step is to make the test tube full of paths that start at the correct city (Mumbai) and end at the correct city (Kolkata). Like we said, there are several operations possible with DNA. Let's use copying. There's an enzyme called polymerase involved. And a reaction called the Polymerase Chain Reaction (PCR). Well, what happens?

PCR can be used to copy sections of DNA you're interested in. The polymerase actually does the copying, and PCR means many iterations of the copying process. You can dictate to the polymerase what segment of DNA you want to start at and where you want it to end. So if we said we wanted it to start at Mumbai and end at Kolkata, those sections of DNA would be copied over and over via PCR. And after a while, you end up with a mix of DNA molecules that encodes several routes, some of them still nonsensical, but all of them starting at Mumbai and ending at Kolkata.

Now we find those paths that are exactly five cities long. Since each city is encoded by four bases, we know the final path should be exactly 20 base pairs long—4 base pairs, multiplied by 5 cities. The technique used to find the length of base pairs is called gel electrophoresis (GE). Using GE, the DNA gets segregated into different lengths, and you can filter out the DNA molecules that have the length you want. So into the next test tube goes all the DNA that is 20 base pairs long. And what does this encode? To reiterate, this new test tube contains all paths that start at Mumbai and end at Kolkata, and which have a total of five cities encoded.